

George Jeff Onyango
+254759059844 | gojeffonyango@gmail.com
| Website: <https://geotasium.co.ke> | P.O Box 0100,
Nairobi, Kenya

PROFESSIONAL SUMMARY

Geoscientist passionate about environmental geoscience, geo-computation, and geochemistry. Proficient in various geophysical tools to collect and model geological data. Specialising in environmental geoscience, hydrogeology, and geochemical modelling for geologic CO₂ sequestration.

EDUCATION

Technical University of Kenya
Master of Technology in Environmental Resource Management

September 2025 - Ongoing

University of Nairobi
Bachelor of Science in Geology

September 2019 - September 2023

WORK EXPERIENCE & PROJECTS

Research Project: Structural and Geochemical Study of Kenyan Rift Basalts for CO₂ Storage

Research Assistant (PI: Lydia Olaka, PhD, Geoscience and The Environment, Technical University of Kenya)

June 2025- July 2025

- Conducted a systematic literature review to synthesise existing knowledge on basaltic petrology, mineral carbonation processes, and CO₂ sequestration methodologies applicable to the Central Kenyan Rift geologic setting.
- Prepared polished thin-sections of basaltic samples and conducted mineralogical (X-ray Diffraction) and elemental (X-ray Fluorescence) analyses to determine mineral assemblages and major/trace element compositions relevant to reactivity and storage potential.
- Integrated geophysical surveying techniques, including magnetic and transient electromagnetic (TDEM) methods to map basalt unit stratigraphy, constrain permeability estimates, and delineate subsurface domains with optimal hydrogeological and structural conditions for CO₂ storage implementation.

Taita Research Station | University of Helsinki

Research Assistant (PI: Morice Odhiambo, PhD)

(Part-time) April 2024 – April 2025

Working as a research assistant as part of Regenerative agricultural systems for climate resilience in agroecological gradient in East Africa (REACT). In the REACT project, the roles include:

- Established experimental irrigation plots and conducted systematic measurements of soil physical properties and crop phenology, including quantification of soil field-saturated hydraulic conductivity using a dual-head infiltrometer (SATURO device).
- Implemented bi-weekly phenological monitoring protocols to assess crop growth dynamics, measuring plant architecture (canopy height), physiological parameters (stomatal conductance via SC-1 Leaf Porometer), root system development, phenological stage, and light interception capacity (Leaf Area Index using Accupar LP-80).
- Investigated soil organic carbon dynamics across contrasting cropping systems and their associations with soil water retention characteristics and hydrological function.

The World Bank & Nomad Link Solution

Hydrogeologist

(Contract) Jan 2025 – Mar 2025

Horn of Africa Groundwater Resilience Project (HOAGW4RP). The project was in collaboration with the World Bank and IGAD, together with the Water Resources Authority in Kenya.

- Conducted geophysical survey using Audio Magnetotellurics, ERT, VES, and Natural Electric Field (PQWT-S Series) to characterise subsurface hydrogeological properties and evaluate aquifer systems in Marsabit and Mandera counties.
- Conducted a borehole inventory survey on both functional and non-functional boreholes to evaluate groundwater resource accessibility and infrastructure status.
- Conducted a multi-criteria evaluation of aquifer potential in Marsabit and Mandera counties to inform site selection for exploratory drilling and the establishment of groundwater monitoring networks.

Scientific and Engineering Power Consultants (SEPCO)

Geology Consultant

(Contract) May 2024 – Jul 2024

- Executed systematic geological field mapping to characterise greenstone belt lithologies, structural geometries, and

- secondary gold mineralisation within tailings deposits, establishing the geological foundation for resource evaluation.
- Collected integrated water and tailings samples across the study area and subjected them to multi-element geochemical analysis to assess environmental contamination profiles and quantify potential environmental impacts associated with legacy mining operations.
- Quantified tailings tonnage and grade distribution within the project domain and evaluated the technical and economic feasibility of establishing a gold recovery facility in Migori County, incorporating site-specific constraints and resource characteristics.
- Conducted comparative techno-economic assessment of Carbon-in-Pulp (CIP) versus Carbon-in-Leach (CIL) metallurgical processes to inform technology selection based on resource characteristics and operational requirements.

Boboro Holdings Investment and Kusini Gateway Industrial Park (KGIP)

Geology Consultant in Songea Beryllium Mines, Tanzania.

Oct 2023 – Jan 2024

- Conducted a comprehensive Environmental Impact Assessment (EIA) consistent with Tanzanian environmental legislation and international best practice standards, employing systematic evaluation protocols to identify, quantify, and mitigate potential ecological consequences of mining operations.
- Implemented systematic quality assurance protocols involving periodic sampling intervals and X-ray Fluorescence (XRF) elemental analysis to verify ore grade conformity with client specifications and maintain analytical standards throughout mining operations.
- Synthesised relevant literature on pegmatite petrogenesis, mineral associations, and extraction methodologies to inform operational practices and identify opportunities for enhanced resource recovery and operational efficiency.

TRAINING & CERTIFICATIONS

- Python Programming and Data Analytics – 365 Data Science (Jul 2021- Feb 2022)
 - Active and Passive Satellite Data Analysis using Cloud Computing for Surface Water/Flood Mapping for Water-Related Application – Water Learning Centre (WLC); United Nations University (2022)
 - Water- Energy- Food (WEF) Nexus Masterclass – IHE Delft & Global Water Partnership (2024)
 - Software Engineering (Backend Development) – Mastercard ALX Africa (2025)
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TECHNICAL SKILLS

- Remote sensing: Proficient in Google Earth Engine, ArcGIS Pro, and QGIS.
- Geophysical methods: Mapping and modelling using Transient Electromagnetic (ABEM WalkTEM 2), VES, ERT & IP (ABEM Terrameter LS2), and Magnetometer (GEM GSM-19).
- Data Analysis using Python, Tableau, and R programming language.

VOLUNTEERING

Centre for Sustainable Mining, Environment, and Climate Change (CESMECC) - Policy and Advocacy Volunteer

- Developing a policy and advocacy framework for mining, ensuring that mining activities are sustainable, socially responsible, and economically beneficial.
- Engaged with local communities to understand their concerns and ensure they benefit from mining activities in Taita Taveta County

PUBLICATIONS

- Onyango, G. J., Adell, M., Ouma, E., & Ogolla, W. (2025).** ASSESSMENT OF MULTI-DECADAL LAKE VICTORIA SHORELINE VARIATIONS USING DIGITAL SHORELINE ANALYSIS SYSTEM (DSAS). *Earth Sciences: An International Journal* . [1125esij08.pdf](#)
- Ouma, E., Adell, M., Ogolla, W., **Onyango, G. J.**, Misongo, M., & Olaka, L. A. (n.d.). Evaluation of geosites in Homabay County, Kenya, as areas of potential geopark development. *Geotourism, Geoparks and Geoheritage* . [Under Review]